

SERVICESUITE CLOUD

API V2

Full Reference

Complete documentation for integrating with the ServiceSuite API V2.
Covers authentication, loans, payments, notifications, fraud analytics, and webhooks.

VERSION

v1

FORMAT

JSON / REST

BASE URL

collectbox.servicesuitecloud.com
/ServiceSuiteApiV2/

SWAGGER UI

.../swagger/index.html

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1. Authentication

All endpoints except `/auth/token` require a **Bearer JWT** in the `Authorization` header. Tokens are scoped to a specific entity — you can only access data belonging to your own organisation.

1.1 Get Access Token PUBLIC

POST `/auth/token`

Generates a JWT access token. Rate-limited to **5 requests per minute**.

Request Body

```
{
  "clientId": "your_client_id",
  "clientSecret": "your_client_secret",
  "entityId": 1
}
```

Field	Type	Required	Description
clientId	string	Yes	API client identifier provided by the admin
clientSecret	string	Yes	API client secret
entityId	integer	Yes	Your organisation / entity ID

200 Success

```
{
  "success": true,
  "message": "Token generated successfully.",
  "data": {
    "accessToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
    "tokenType": "Bearer",
    "expiresIn": 3600
  }
}
```

Include the token in every subsequent request: `Authorization: Bearer <accessToken>`

1.2 Create API Client ADMIN ONLY

POST `/auth/clients`

Provisions a new API client. Requires the `X-Admin-Key` header. Not for use by integrators — contact the ServicesSuite team.

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Generated June 2026

Important: The `clientSecret` is only returned once. Store it securely immediately — it cannot be retrieved again.

2. Loans

All loan endpoints require `BEARER JWT`. The `entityId` is automatically derived from your token — you can only access your own entity's data.

2.1 Get Loans by Filter `JWT`

POST

`/loans/GetLoansByFilter`

Returns active loans matching the supplied filter criteria. All filter fields are optional — omit any field to skip that filter.

Request Body

```
{
  "minDays": 30,
  "maxDays": 90,
  "minAmount": 5000,
  "maxAmount": 100000,
  "minOlb": 1000,
  "maxOlb": 50000
}
```

Field	Type	Description
<code>minDays</code>	integer?	Minimum days past due
<code>maxDays</code>	integer?	Maximum days past due
<code>minAmount</code>	decimal?	Minimum original disbursed amount
<code>maxAmount</code>	decimal?	Maximum original disbursed amount
<code>minOlb</code>	decimal?	Minimum outstanding loan balance
<code>maxOlb</code>	decimal?	Maximum outstanding loan balance

200 Success — returns `LoanResponse` (count + array of `LoanDto`)

2.2 Get Active Loans `JWT`

GET

`/loans/ActiveLoans`

Returns a list of active loans (balance > 0) with optional keyword search.

Query Param	Type	Required	Default	Description
<code>search</code> CONFIDENTIAL — For authorised integrators only	string	No	—	Filter by borrower name, phone, or loan ID Generated June 2026
<code>top</code>	integer	No	<code>20</code>	Maximum records to return

```
GET /loans/ActiveLoans?search=0712345678&top=50
```

2.3 Get Loan by ID

JWT

GET

/loans/{loanId}

GET

/loans/GetLoanById/{loanId}

Returns a single loan by its numeric ID. Returns **404** if the loan does not exist or belongs to another entity.

```
GET /loans/4501
```

200 Success

```
{
  "success": true,
  "message": "Loan retrieved successfully.",
  "data": { /* LoanDto */ }
}
```

2.4 Get Loan Details

JWT

GET

/loans/details/{loanId}

Returns detailed form/metadata items attached to a loan (custom fields from borrower onboarding forms).

200 Success

```
{
  "success": true,
  "data": [
    { "id": 1, "loanId": "4501", "itemName": "Employer Name", "itemValue": "ABC Company Ltd" }
  ]
}
```

2.5 Get Loan Balance

JWT

GET

/loans/balance/{loanId}

Returns the current outstanding balance for a single loan.

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200 Success

```
{
  "success": true,
  "message": "Balance retrieved successfully.",
  "data": { "id": "4501", "loanBalance": 32000.00 }
}
```

2.6 Get Disbursed Loans JWT

GET `/loans/disbursements`

Returns all loans disbursed within a date range, ordered by disbursement date descending.

Query Param	Type	Required	Description
<code>startDate</code>	datetime	Yes	Start of range e.g. <code>2024-01-01</code>
<code>endDate</code>	datetime	Yes	End of range e.g. <code>2024-01-31</code>

200 Success

```
{
  "success": true,
  "message": "5 disbursement(s) found.",
  "data": [
    {
      "loanId": "4501", "borrowerName": "Jane Wanjiru",
      "phoneNumber": "0712345678", "amountToDisburse": 50000.00,
      "disbursementDate": "2024-01-15T00:00:00", "currentBalance": 32000.00
    }
  ]
}
```

2.7 Get Payments JWT

GET `/loans/payments`

Returns all payments received within a date range (from the Transactions database), ordered by date descending.

Query Param	Type	Required	Description
<code>startDate</code>	datetime	Yes	Start of range
<code>endDate</code>	datetime	Yes	End of range

200 Success

```
{
  "data": [
    {
      "id": 1, "transId": "QHX7YABCDE", "transAmount": 2000.00,
      "billRefNumber": "4501", "payerName": "Jane Wanjiru",
      "dateDone": "2024-01-20T10:34:00", "isPosted": 1,
      "transactionType": "Pay Bill", "loanId": "4501"
    }
  ]
}
```

2.8 Get Overdue Loans

JWT

GET

/loans/overdue

Returns loans that are past their due date, ordered by days in arrears descending.

Query Param	Type	Required	Default	Description
minDays	integer	No	1	Minimum days past due
top	integer	No	50	Maximum records to return

2.9 Get Loans Due Today

JWT

GET

/loans/due-today

Returns loans that have an installment scheduled for today (unpaid), ordered by amount due descending.

Query Param	Type	Required	Default	Description
top	integer	No	500	Maximum records to return

200 Sample record

```
{
  "loanId": "4501", "firstName": "Jane", "otherName": "Wanjiru",
  "phoneNumber": "0712345678", "loanBalance": 32000.00,
  "dueTodayAmount": 5000.00, "productName": "Monthly Loan",
  "dueDate": "2024-06-16T00:00:00"
}
```


2.10 Get Borrower

JWT

GET

/loans/borrower

Looks up a client's personal details by phone number, National ID, or borrower ID.

Query Param	Type	Required	Description
search	string	Yes	Phone number, National ID, or Borrower ID

200 Success

```
{
  "data": {
    "borrowerId": "101", "firstName": "Jane", "otherName": "Wanjiru",
    "nationalID": "30123456", "phoneNumber": "0712345678",
    "emailAddress": "jane@email.com", "accountNo": "ACC-00101",
    "accountStatus": 1
  }
}
```

2.11 Get Borrower Loans

JWT

GET

/loans/borrower/loans

Returns all loans (active and historical) for a specific borrower.

Query Param	Type	Required	Description
search	string	Yes	Phone number, National ID, or Borrower ID

2.12 Get Borrower Statement

JWT

GET

/loans/borrower/statement

Returns the full transaction statement for a borrower — all disbursements, repayments, and adjustments — ordered chronologically.

Query Param	Type	Required	Description
search	string	Yes	Phone number, National ID, or Borrower ID

```
{
  "data": {
    "borrowerId": "101", "borrowerName": "Jane Wanjiru",
    "totalLines": 24,
    "statement": [
      {
        "id": 1, "loanId": "4501", "amount": 50000.00,
        "transType": 1, "narration": "Loan Disbursement",
        "mpesaRef": "", "loanBalance": 50000.00,
        "transactedDate": "2024-01-15T00:00:00"
      }
    ]
  }
}
```

transType	Meaning
1	Disbursement
2	Repayment
3	Penalty
6	Write-off

2.13 Get Client Profile — Combined

JWT

GET

/loans/borrower/profile

Returns a client's personal details **and** all their currently active loans in a single call. Best used for client lookup screens.

Query Param	Type	Required	Description
search	string	Yes	Phone number, National ID, or Borrower ID

```
{
  "success": true,
  "message": "2 active loan(s) found.",
  "data": {
    "client": {
      "borrowerId": "101", "firstName": "Jane", "otherName": "Wanjiru",
      "nationalID": "30123456", "phoneNumber": "0712345678",
      "emailAddress": "jane@email.com", "accountNo": "ACC-00101",
      "accountStatus": 1
    },
    "activeLoans": [
      {
        "id": "4501", "amountToDisburse": 50000.00,
        "loanBalance": 32000.00, "arrears": 5000.00,
        "daysInArrears": 15, "productName": "Monthly Loan",
        "agent": "John Kamau"
      }
    ]
  }
}
```

activeLoans only includes loans where `LoanBalance > 0`. Returns an empty array `[]` if the client has no outstanding loans.

3. Payments

3.1 Initiate M-Pesa STK Push JWT

POST `/payments/stk-push`

Triggers an M-Pesa STK push prompt on the customer's phone. M-Pesa credentials are configured per entity in the system database.

Request Body

```
{
  "amount":      2000,
  "phoneNumber": "254712345678",
  "accountReference": "4501",
  "transactionDesc": "Loan Repayment"
}
```

Field	Type	Required	Description
amount	decimal	Yes	Amount to charge. Must be > 0
phoneNumber	string	Yes	International format: 254XXXXXXXXX
accountReference	string	No	Reference shown on the STK prompt (e.g. loan ID)
transactionDesc	string	No	Description on the prompt. Defaults to "Payment"

200 Success

```
{
  "success": true,
  "message": "STK push initiated.",
  "data":    "ws_CO_16062024103400001234567890"
}
```

data contains the M-Pesa CheckoutRequestID. Use it to reconcile the payment when the M-Pesa callback arrives.

4. Notifications

4.1 Send SMS

JWT

POST

/notifications/sms

Sends an SMS to the specified phone number using the entity's configured SMS provider.

Request Body

```
{
  "message": "Dear Jane, your loan balance is KES 32,000. Please pay to avoid penalties.",
  "phoneNumber": "0712345678",
  "scheduleDate": "2024-06-16T10:00:00"
}
```

Field	Type	Required	Description
message	string	Yes	SMS body text
phoneNumber	string	Yes	Recipient phone number
scheduleDate	datetime	No	Future datetime for scheduled delivery. Omit to send immediately

5. Fraud Analytics

All fraud endpoints require **BEARER JWT** and are scoped to your entity. Risk levels: **High** | **Medium** | **Low**.

5.1 Full Fraud Report JWT

GET `/analytics/fraud/report`

Runs all fraud signal queries in parallel and returns everything in one response. Use this for dashboard initial load.

200 Response shape

```
{
  "data": {
    "summary": {
      "highRiskAgentCount": 2, "highRiskShopCount": 1,
      "duplicateIdentityCount": 3, "loanStackingCount": 5,
      "zeroPaymentLoanCount": 8, "estimatedFraudExposure": 320000.00,
      "generatedAt": "2024-06-16T10:00:00Z"
    },
    "agentSignals": [ /* AgentFraudSignal[] */ ],
    "shopSignals": [ /* ShopFraudSignal[] */ ],
    "duplicateIdentities": [ /* DuplicateIdentitySignal[] */ ],
    "loanStacking": [ /* LoanStackingSignal[] */ ],
    "suspiciousLoans": [ /* SuspiciousLoanSignal[] */ ],
    "paymentFraud": { "unpostedCount": 2, "unpostedAmount": 15000.00 }
  }
}
```

5.2 Agent Fraud Signals JWT

GET `/analytics/fraud/agents`

Flags collection agents with high default rates, write-off concentration, or arrears-heavy portfolios.

200 Sample record

```
{
  "agentId": "12", "agentName": "John Kamau",
  "totalLoans": 50, "defaultedLoans": 20, "defaultRate": 0.40,
  "totalArrears": 200000.00, "writeoffCount": 5,
  "riskLevel": "High",
  "flags": [ "High default rate (40%)", "Concentrated write-offs" ]
}
```

5.3 Shop / Merchant Fraud Signals

JWT

GET

/analytics/fraud/shops

Groups borrowers by onboarding agent (shop) and flags shops where the majority of clients are defaulting.

200 Sample record

```
{
  "agentName": "John Kamau Shop", "uniqueBorrowers": 25,
  "defaultedBorrowers": 20, "borrowerDefaultRate": 0.80,
  "totalDisbursed": 250000.00, "portfolioAtRisk": 0.72,
  "riskLevel": "High"
}
```

5.4 Duplicate Identity Detection

JWT

GET

/analytics/fraud/borrowers/duplicate-ids

Identifies borrowers sharing the same National ID — a signal of synthetic or stolen identity fraud.

200 Sample record

```
{
  "nationalId": "30123456", "borrowerCount": 2,
  "names": "Jane Wanjiru, Jane W Mwangi",
  "phoneNumbers": "0712345678, 0798765432",
  "totalActiveBalance": 82000.00
}
```

5.5 Loan Stacking Detection

JWT

GET

/analytics/fraud/borrowers/stacking

Identifies borrowers holding more than one active loan simultaneously — over-borrowing or coordinated fraud.

200 Sample record

```
{
  "borrowerName": "Jane Wanjiru", "phoneNumber": "0712345678",
  "activeLoanCount": 2, "totalBalance": 82000.00,
  "firstLoanDate": "2023-10-01T00:00:00", "latestLoanDate": "2024-01-15T00:00:00"
}
```

5.6 Suspicious Loans

JWT

GET

/analytics/fraud/loans/suspicious

Returns overdue loans with little or no repayment activity. Zero-payment loans appear first.

Query Param	Type	Required	Default	Description
minDays	integer	No	30	Minimum days overdue

200 Sample record

```
{
  "loanId": "4501", "borrowerName": "Jane Wanjiru",
  "amountDisbursed": 50000.00, "daysInArrears": 152,
  "paymentCount": 0, "totalPaid": 0.00
}
```

5.7 Unposted Payment Fraud

JWT

GET

/analytics/fraud/payments/unposted

Flags payments received but not posted to a loan for 3+ days — potential staff diversion of funds.

200 Success

```
{
  "data": {
    "unpostedCount": 2,
    "unpostedAmount": 15000.00,
    "oldestUnposted": "2024-06-10T00:00:00",
    "newestUnposted": "2024-06-13T00:00:00"
  }
}
```


6. Webhooks

6.1 Spin Analysis Webhook BASIC AUTH

POST

/webhooks/spin-analysis

Receives an analysis result from the Spin financial analysis service. Uses HTTP Basic Auth — **not** Bearer JWT.

Authorization: Basic <base64(username:password)>

Webhook credentials are configured server-side. Contact the ServiceSuite team to obtain the username and password.

Request Body

```
{
  "file_unique_id": "abc123xyz",
  "file_type":      "mpesa",
  "phone":          "0712345678",
  "id_number":      "30123456",
  "duration":       6.0,
  "json_data":      {},
  "timestamp":      "2024-06-16T10:00:00Z",
  "state_name":     "completed"
}
```

Field	Type	Description
file_unique_id	string	Unique ID of the analysed file
file_type	string	"mpesa" or "bank"
phone	string?	Phone number (M-Pesa statements)
id_number	string?	National ID
bank_name	string?	Bank name (bank statements)
account_number	string?	Bank account number
duration	decimal?	Statement period in months
json_data	object?	Raw analysis output from Spin
timestamp	datetime?	Time the analysis completed
state_name	string?	e.g. "completed" , "failed"

7. Schema Reference

Response Envelope — ApiResponse<T>

Field	Type	Description
success	boolean	true on success, false on error
message	string	Human-readable status message
data	T null	Response payload. null on error

LoanDto

Field	Type	Description
id	string	Loan ID
borrowerId	string	Borrower ID
firstName	string	Borrower first name
otherName	string	Borrower other name
phoneNumber	string	Borrower phone number
emailAddress	string	Borrower email
nationalId	string	National ID
amountToDisburse	decimal	Original disbursed amount
repaymentPeriod	string	Loan product name
loanBalance	decimal	Current outstanding balance
penalty	decimal	Accrued penalty
arrears	decimal	Total overdue amount
daysInArrears	integer	Days past due
outsourcedAmount	decimal	Sum of all unpaid installments
productName	string	Loan product
agent	string	Collection agent full name
agentId	string	Collection agent ID

BorrowerDto

Field	Type	Description
borrowerId	string	Unique borrower ID
firstName	string	First name
otherName	string	Other name / surname
nationalID	string	National ID
phoneNumber	string	Phone number
emailAddress	string	Email address
accountNo	string	Account number
accountStatus	integer	1 = Active, 0 = Inactive

PaymentDto

Field	Type	Description
id	integer	Payment record ID
transId	string	M-Pesa transaction ID
transAmount	decimal	Payment amount
billRefNumber	string	Bill reference (loan account)
payerName	string	Name of the payer
dateDone	datetime	Payment date
isPosted	integer	1 = posted, 0 = pending
transactionType	string	e.g. "Pay Bill"
loanId	string?	Associated loan ID

8. Error Reference

Status	Meaning	When it occurs
400	Bad Request	Missing required field or invalid parameter value
401	Unauthorized	Missing, expired, or invalid token; wrong Basic Auth credentials
404	Not Found	Resource does not exist or belongs to a different entity
429	Too Many Requests	Rate limit exceeded on <code>/auth/token</code> (5 req / min)
500	Internal Server Error	Unexpected server-side error — check server logs

All error responses follow the same envelope:

```
{
  "success": false,
  "message": "Description of what went wrong.",
  "data": null
}
```

9. Code Examples

cURL — Full Flow

```
# Step 1 — Get token
TOKEN=$(curl -s -X POST \
  https://collectbox.servicesuitecloud.com/ServiceSuiteApiV2/auth/token \
  -H "Content-Type: application/json" \
  -d '{"clientId":"your_id","clientSecret":"your_secret","entityId":1}' \
  | jq -r '.data.accessToken')

# Step 2 — Client profile
curl -X GET \
  "https://collectbox.servicesuitecloud.com/ServiceSuiteApiV2/loans/borrower/profile?search=0712345678" \
  -H "Authorization: Bearer $TOKEN"

# Step 3 — STK push
curl -X POST \
  https://collectbox.servicesuitecloud.com/ServiceSuiteApiV2/payments/stk-push \
  -H "Authorization: Bearer $TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"amount":2000,"phoneNumber":"254712345678","accountReference":"4501"}'

# Step 4 — Fraud report
curl -X GET \
  https://collectbox.servicesuitecloud.com/ServiceSuiteApiV2/analytics/fraud/report \
  -H "Authorization: Bearer $TOKEN"
```

JavaScript (fetch)

```
const BASE = "https://collectbox.servicesuitecloud.com/ServiceSuiteApiV2";

async function getToken() {
  const res = await fetch(`${BASE}/auth/token`, {
    method: "POST",
    headers: { "Content-Type": "application/json" },
    body: JSON.stringify({ clientId: "your_id", clientSecret: "your_secret", entityId: 1 })
  });
  const { data } = await res.json();
  return data.accessToken;
}

async function apiGet(token, path) {
  return fetch(`${BASE}${path}`, {
    headers: { Authorization: `Bearer ${token}` }
  }).then(r => r.json());
}

// Usage
const token = await getToken();
const profile = await apiGet(token, "/loans/borrower/profile?search=0712345678");
const overdue = await apiGet(token, "/loans/overdue?minDays=30");
const fraud = await apiGet(token, "/analytics/fraud/report");
```

C# (HttpClient)

```
using var http = new HttpClient {
  BaseAddress = new Uri("https://collectbox.servicesuitecloud.com/ServiceSuiteApiV2/")
};

// Authenticate
var auth = await http.PostAsJsonAsync("auth/token",
  new { clientId = "your_id", clientSecret = "your_secret", entityId = 1 });
var token = (await auth.Content.ReadFromJsonAsync<ApiResponse<TokenResponse>>())!.Data!.AccessToken;
http.DefaultRequestHeaders.Authorization = new("Bearer", token);

// Call endpoints
var profile = await http.GetFromJsonAsync<ApiResponse<ClientProfileDto>>(
  "loans/borrower/profile?search=0712345678");
var fraud = await http.GetFromJsonAsync<ApiResponse<FraudAnalyticsReport>>(
  "analytics/fraud/report");
```

PHP (cURL)

```
<?php
$base = "https://collectbox.servicesuitecloud.com/ServiceSuiteApiV2";

function apiPost($base, $token, $path, $body) {
    $ch = curl_init("$base$path");
    curl_setopt_array($ch, [
        CURLOPT_POST          => true,
        CURLOPT_RETURNTRANSFER => true,
        CURLOPT_HTTPHEADER    => ["Authorization: Bearer $token", "Content-Type: application/json"],
        CURLOPT_POSTFIELDS     => json_encode($body)
    ]);
    $r = curl_exec($ch); curl_close($ch);
    return json_decode($r, true);
}

function apiGet($base, $token, $path) {
    $ch = curl_init("$base$path");
    curl_setopt_array($ch, [CURLOPT_RETURNTRANSFER => true,
        CURLOPT_HTTPHEADER => ["Authorization: Bearer $token"]]);
    $r = curl_exec($ch); curl_close($ch);
    return json_decode($r, true);
}

$auth = apiPost($base, "", "/auth/token",
    ["clientId=>"your_id", "clientSecret">"your_secret", "entityId">1]);
$token = $auth["data"]["accessToken"];

$profile = apiGet($base, $token, "/loans/borrower/profile?search=0712345678");
$fraud = apiGet($base, $token, "/analytics/fraud/report");
```